

Implement a Sobel algorithm using Open CV to filter the input image.

Code:

```
import cv2, urllib.request

from google.colab.patches import cv2_imshow

# Download sample image

urllib.request.urlretrieve(

    "https://raw.githubusercontent.com/opencv/opencv/master/samples/data/lena.jpg",

    "img.jpg"

)

# Read image

img = cv2.imread("img.jpg")

# Convert to grayscale

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Apply Sobel filter

sobelx = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)

sobely = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)

# Display images

print("Sobel X")

cv2_imshow(sobelx)

print("Sobel Y")

cv2_imshow(sobely)
```

Output:



sobel Y

